Homework 1

Due : Friday, February 20 at 11 : 59 pm

Deliverables. Submit a single PDF of your write-up to Gradescope *HW1 Write-Up*.

Start each problem on a new page.

Honor Code

Write and sign the following statement:

I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted. James Dinh

"I certify that all solutions in this document are entirely my own and that I have not looked at anyone else's solution. I have given credit to all external sources I consulted."

Linear Algebra

1. **System of equations (3 points).** Consider the linear system

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}.$$

How many solutions does this system have? Explain your reasoning.

$$\begin{bmatrix} 1 & 2 & 3 & 6 \\ 2 & 0 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow[R_3-3R_1]{R_2-2R_1} \begin{bmatrix} 1 & 2 & 3 & 6 \\ 0 & -4 & 0 & -4 \\ 0 & 0 & 0 & -6 \end{bmatrix} \xrightarrow{R_2/-4} \begin{bmatrix} 1 & 2 & 3 & 6 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & -6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Row Reduce

$$\Rightarrow \begin{cases} x + 2y + 3z = 6 \\ y = 1 \\ \underbrace{0 = -6}_{\text{contradiction}} \end{cases} \Rightarrow \boxed{\text{no solution}}$$

This system of equations has no solution because there is a contradiction with the first and third rows. While row 3 is a scalar of row 1, the solutions are inconsistent, leaving us with $0=-6$ after row reduction.

2. **Asymptotic powers of 2×2 matrices (6 points).** For each matrix below, determine the behavior of

$$\lim_{n \to \infty} M^n$$

Hint: Use the eigen-decomposition $M = PDP^{-1}$, where $D = \{\lambda_1, \lambda_2, ...\}$ is a diagonal matrix of the eigenvalues. What would be the formula for M^n?

(a) $M_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

(b) $M_2 = \begin{bmatrix} 2 & -5 \\ 1/2 & -7/6 \end{bmatrix}$.

(c) $M_3 = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{n=2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{n=3}$$

a) M_1: $\det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = 0$

$\lambda^2 - 1 = 0$

$(\lambda - 1)(\lambda + 1) = 0$

$\lambda = 1, -1$

e-vectors

$\lambda = 1: \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-x + y = 0$
$x - y = 0$ $\Rightarrow \{ x = y$

e-space: $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

$\lambda = -1: \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$x + y = 0 \Rightarrow x = -y$

e-space: $\text{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$

$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $D^n = \begin{bmatrix} 1^n & 0 \\ 0 & (-1)^n \end{bmatrix}$

$M_1 = PDP^{-1}$ $M_1^2 = PD\cancel{P^{-1}P}DP^{-1} = PD^2P^{-1}$ $M_1^n = PD^nP^{-1}$

Because in our diagonal eigenvalue matrix, we have an eigenvalue that oscillates between 1 and -1 depending on n, the limit of M_1^n as $n \to \infty$ $\boxed{\text{does not exist.}}$

b) M_2: $\det \begin{bmatrix} 2-\lambda & -5 \\ 1/2 & -7/6 - \lambda \end{bmatrix} = 0$

$(2-\lambda)(-\frac{7}{6} - \lambda) + \frac{5}{2} = 0$

$-\frac{7}{3} - 2\lambda + \frac{7}{6}\lambda + \lambda^2 + \frac{5}{2} = 0$

$-\frac{14}{6} + \frac{15}{6}$

$\lambda^2 - \frac{5}{6}\lambda + \frac{1}{6} = 0$

$(\lambda - \frac{3}{6})(\lambda - \frac{2}{6}) = 0$

$\lambda = \frac{1}{2}, \frac{1}{3}$

e-vectors

$\begin{bmatrix} 2-\frac{1}{2} & -5 \\ 1/2 & -\frac{7}{6} - \frac{1}{2} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\frac{3}{2}x - 5y = 0$
$\frac{1}{2}x - \frac{10}{6}y = 0$ $\Rightarrow \{ x = \frac{10}{3}y$

e-space: $\text{span}\left\{\begin{pmatrix} 10/3 \\ 1 \end{pmatrix}\right\}$

$\begin{bmatrix} 2-\frac{1}{3} & -5 \\ 1/2 & -\frac{7}{6} - \frac{1}{3} \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\frac{5}{3}x - 5y = 0$
$\frac{1}{2}x - \frac{9}{6}y = 0$ $\Rightarrow \{ x = 3y$

e-space: $\text{span}\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}\right\}$

$M_2^n = \begin{bmatrix} 10/3 & 3 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} (1/2)^n & 0 \\ 0 & (1/3)^n \end{bmatrix}\begin{bmatrix} 10/3 & 3 \\ 1 & 1 \end{bmatrix}^{-1}$

$D^n = \begin{bmatrix} 1/2^n & 0 \\ 0 & 1/3^n \end{bmatrix}$

$\lim_{n \to \infty} M_2^n = \lim_{n \to \infty} D^n = \boxed{0}$

$\hookrightarrow$ both $\frac{1}{2^n}$ and $\frac{1}{3^n}$ approach 0 as $n \to \infty$

c) M_3: $\det\begin{bmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{bmatrix} = 0$

$(4-\lambda)(1-\lambda)+2 = 0$

$\lambda^2 - 5\lambda + 6 = 0$

$(\lambda-2)(\lambda-3) = 0$

$\lambda = 2, 3$

$D_3^n = \begin{bmatrix} 2^n & 0 \\ 0 & 3^n \end{bmatrix}$

approaches ∞

$\lim_{n\to\infty} D_3^n = \begin{bmatrix} 2^\infty & 0 \\ 0 & 3^\infty \end{bmatrix}$

Thus, $\boxed{\lim_{n\to\infty} M_3^n \text{ diverges and DNE}}$

3. **Singular Value Decomposition (4 points).** Given the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$$

Find a unit vector x ($\|x\| = 1$) for which $\|Ax\|$ is maximized.

Hint: Recall from SVD that $A = U\Sigma V^\top$, where A maps each right singular vector (a column of V) to a left singular vector (a column of U), scaled by the corresponding singular value. For which right singular vector does this scaling make $\|Ax\|$ the largest?

$$A^T = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix} \qquad A^TA = \begin{bmatrix} 2 & 2 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix}$$

evals:

$$\det\left(\begin{bmatrix} 8-\lambda & 2 \\ 2 & 5-\lambda \end{bmatrix}\right) = 0$$

$$(8-\lambda)(5-\lambda) - 4 = 0$$

$$\lambda^2 - 13\lambda + 36 = 0$$

$$(\lambda-9)(\lambda-4) = 0$$

$$\lambda = 9, 4$$

$$\sigma_1 = 3 \qquad \sigma_2 = 2$$

3 is largest singular value

evectors: $\lambda = 9$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x + 2y = 0 \Rightarrow x = 2y$$
$$2x - 4y = 0$$

$$\text{span}\left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$$

normalize: $\|v\| = \sqrt{4+1} = \sqrt{5}$

$$x = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$\therefore$ the unit vector x where $\|Ax\|$ is maximized is $\boxed{\frac{1}{\sqrt{5}}\begin{bmatrix} 2 \\ 1 \end{bmatrix}}$

4. **Image Flipping (6 points).** *This question is a warmup to problem 4 in the coding section.* Consider a tiny 3×3 image represented by matrix I and the same image flipped on the Y axis I_{flip}:

$$I = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \\ x_7 & x_8 & x_9 \end{bmatrix}, \quad I_{\text{flip}} = \begin{bmatrix} x_7 & x_8 & x_9 \\ x_4 & x_5 & x_6 \\ x_1 & x_2 & x_3 \end{bmatrix}.$$

   (a) What is the size of the transformation matrix T that performs this flip? *Hint: You can first convert I to a 1×9 vector, make transformation using matrix T, and then convert I_{flip} back to a 3×3 matrix.*

   (b) Construct T and verify that $I \times T$ produces I_{flip}.

   (c) Describe an algorithm for constructing the vertical-flip transformation matrix for any $N \times N$ matrix (either text explanation or pseudocode is acceptable). Show that $I \times T$ produces I_{flip}.

   (d) Describe how you would modify this algorithm to do a *horizontal* flip (along the X ais).

$I = (x_1, \dots, x_9)$

$I_{flip} = (x_7, x_8, x_9, x_4, x_5, x_6, x_1, x_2, x_3)$

a) Since we can convert I to a 1×9 vector, then transform it to an output vector that is also 1×9, T is a $\boxed{9 \times 9}$ matrix.

b)
$$T = \begin{bmatrix} 0&0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&0&0&0&1 \\ 0&0&0&1&0&0&0&0&0 \\ 0&0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&1&0&0&0 \\ 1&0&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0&0 \end{bmatrix}$$

$IT = [x_1 \cdots x_9]\, T = [x_7\ x_8\ x_9\ x_4\ x_5\ x_6\ x_1\ x_2\ x_3]$ $\checkmark$

linear index: $k = (i-1)N + j$

flip row index: $i' = N - i + 1$ (by definition)

substitute in linear index: $k' = ((N-i+1)-1)N + j$

$= (N-i)N + j$

c) $T = [N][N]$ matrix

```
for i = 1 : N:
    for j = 1 : N:
        K = (i-1)*N + j     # this gets the linear index for the row
        K' = (N-i)*N + j    # this gets the flipped linear index
        T(K, K') = 1
```

$I \times T = I_{flip}$

$(vT)_{k'} = v_k$

$(vT)_{(N-i)N+j} = I_{ij}$

$(vT)_{(i',j)} = I_{(N-i'+1),j}$

d) To modify this algorithm for a horizontal flip, we would need to apply the flip operation to j instead of i,

$j' = N - j + 1 \Rightarrow k' = (i-1)N + (N-j+1)$

Calculus

5. **Partial Derivatives (8 points). (First and second derivatives)**

   (a) For $f(x_1, x_2) = x_1^3 + x_2^3 - 3x_1x_2$, find all the first and second order partial derivatives.

   (b) Let $f(x, y) = 4x^2 + y^2 - 8xy + 4x + 6y - 10$.
   Find the critical points by solving

   $$\frac{\partial f}{\partial x} = 0 \quad \text{and} \quad \frac{\partial f}{\partial y} = 0$$

   simultaneously and determine which point(s) yield a minimum value.

   (c) For $f(x, y) = e^{xy} + x^2y$, compute:
   (i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$
   (ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

   (d) For $f(x, y) = \ln(x^2 + y^2 + 1)$, compute:
   (i) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$
   (ii) $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y^2}$, and $\frac{\partial^2 f}{\partial x \partial y}$

a) 1st $\frac{\partial}{\partial x_1}(x_1^3 + x_2^3 - 3x_1x_2) = 3x_1^2 - 3x_2$

2nd $\frac{\partial}{\partial x_1}(3x_1^2 - 3x_2) = 6x_1$

1st $\frac{\partial}{\partial x_2}(x_1^3 + x_2^3 - 3x_1x_2) = 3x_2^2 - 3x_1$

2nd $\frac{\partial}{\partial x_2}(3x_2^2 - 3x_1) = 6x_2$

b) $\frac{\partial}{\partial x}(4x^2 + y^2 - 8xy + 4x + 6y - 10) = 8x - 8y + 4 = 0$

$\frac{\partial}{\partial y}(4x^2 + y^2 - 8xy + 4x + 6y - 10) = 2y - 8x + 6 = 0$

$\Rightarrow \begin{cases} -6y + 10 = 0 \\ y = \frac{5}{3} \end{cases}$

$\frac{10}{3} + \frac{18}{3} = 8x$

$8x = \frac{28}{3}$

$x = \frac{7}{6}$

crit point: $\left(\frac{7}{6}, \frac{5}{3}\right)$

Hessian: $\begin{pmatrix} 8 & -8 \\ -8 & 2 \end{pmatrix}$

$\det(H) = 16 - 64 = \underline{-48 < 0}$ $\therefore$ $\boxed{\left(\frac{7}{6}, \frac{5}{3}\right) \text{ is a saddle point and there is no minimum}}$

c) $f(x,y) = e^{xy} + x^2 y$

(i) $\dfrac{\partial f}{\partial x} = ye^{xy} + 2xy$ $\qquad \dfrac{\partial f}{\partial y} = xe^{xy} + x^2$

(ii) $\dfrac{\partial^2 f}{\partial x^2} = y^2 e^{xy} + 2y$ $\qquad \dfrac{\partial^2 f}{\partial y^2} = x^2 e^{xy}$ $\qquad \dfrac{\partial^2 f}{\partial x \partial y} = e^{xy} + xye^{xy} + 2x$

d) $f(x,y) = \ln(x^2+y^2+1)$

(i) $\dfrac{\partial f}{\partial x} = \dfrac{2x}{x^2+y^2+1}$ $\qquad \dfrac{\partial f}{\partial y} = \dfrac{2y}{x^2+y^2+1}$

(ii) $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{2(x^2+y^2+1) - 4x^2}{(x^2+y^2+1)^2}$ $\qquad \dfrac{\partial^2 f}{\partial y^2} = \dfrac{2(x^2+y^2+1) - 4y^2}{(x^2+y^2+1)^2}$ $\qquad \dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{2(x^2+y^2+1) - 4xy}{(x^2+y^2+1)^2}$

6. **Recursive expression and derivatives (6 points).** Suppose we have variables $z_1, z_2, \ldots, z_n$, $w_1, \ldots, w_{n-1}$, and $b_1, \ldots, b_{n-1} \in \mathbb{R}$, where

$$z_n = w_{n-1} z_{n-1} + b_{n-1}.$$

(a) Compute $\dfrac{dz_k}{dz_{k-1}}$ for $k = 2, \ldots, n$.

(b) Compute $\dfrac{dz_n}{dz_1}$.

(c) Compute $\dfrac{dz_n}{db_1}$. You may express your answers in terms of $z_1, \ldots, z_n, w_1, \ldots, w_{n-1}, b_1, \ldots, b_{n-1}$

a) $\dfrac{dz_K}{dz_{K-1}} = \boxed{W_{K-1}}$

b) $\dfrac{dz_n}{dz_1} = \dfrac{dz_n}{dz_{n-1}} \cdot \dfrac{dz_{n-1}}{dz_{n-2}} \cdot \cdots \cdot \dfrac{dz_2}{dz_1} = w_{n-1} \cdot w_{n-2} \cdot \cdots \cdot w_1 = \boxed{\prod_{K=1}^{n-1} w_K}$

c) $\dfrac{dz_n}{db_1} = \dfrac{dz_n}{dz_{n-1}} \cdot \dfrac{dz_{n-1}}{dz_{n-2}} \cdot \cdots \cdot \dfrac{dz_3}{dz_2} \cdot \dfrac{dz_2}{db_1} = w_{n-1} \cdot w_{n-2} \cdot \cdots \cdot w_2 \cdot \dfrac{d}{db_1}(w_1 z_1 + b_1)$

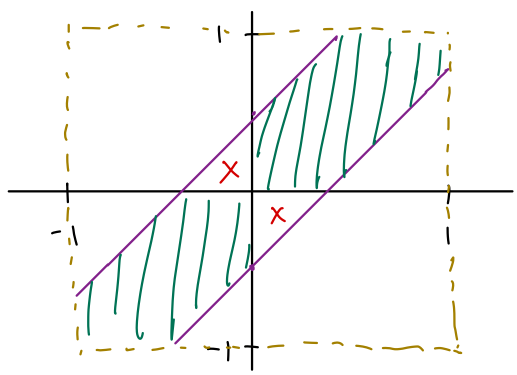
$= \prod_{K=2}^{n-1} w_K \cdot 1 = \boxed{\prod_{K=2}^{n-1} w_K}$

Probability

7. **Conditioned uniform difference (3 points).** Let $X, Y \overset{\text{iid}}{\sim} \text{Unif}(-1, 1)$. Compute

$$\mathbb{P}\big(|X - Y| \le 0.5 \,\big|\, X\,Y > 0\big).$$

Hint: Try drawing a 2D cartesian plane where the horizontal and vertical axes represent X and Y respectively and each range from -1 to 1. For which quadrants is it true that $XY > 0$? Within these quadrants, how can we visualize the region $|X - Y| < 0.5$?



$|X - Y| < 0.5$

$\begin{cases} X - Y < 0.5 \\ X - Y < -0.5 \end{cases} \Rightarrow \begin{matrix} Y < X - 0.5 \\ Y < X + 0.5 \end{matrix}$

$XY > 0$ in first and third quadrant

$P(|X - Y| \le 0.5 \mid XY > 0)$

$\hookrightarrow P(XY > 0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Area of 1 quadrant: $1 - 2\left(\frac{0.5^2}{2}\right) = 1 - 0.25 = 0.75$

2 quadrants: $2 \times 0.75 = 1.5$

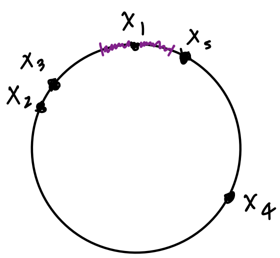
$\Rightarrow \dfrac{\text{Area of } |X-Y| \le 0.5}{\text{Area of } XY > 0} = \dfrac{1.5}{2} = \boxed{0.75}$

8. **Nearest–neighbor arc length (4 points).** Suppose you select 20 i.i.d. points $X_1, \ldots, X_{20}$ uniformly at random on the circumference of the unit circle.

   (a) Let D be the shortest arc distance from X_1 to the nearest of the other 19 points. Calculate $\mathbb{P}(D > t)$ where $0 \le t \le \frac{1}{2}$.
   Hint: What does the event $\{D > t\}$ mean in terms of where the other 19 points can be located relative to X_1?

   (b) Find the expected arc length (in degrees) between X_1 and the point nearest to it.
   Hint: The Wikipedia page on expected value of continuous variables *may be helpful here.*

a)



$P(D > t)$ = prob that all of other 19 points are outside the distance t around X_1

↳ $P(X_2 \text{ not in } X_1 \text{ boundary}) = \dfrac{2\pi - 2t}{2\pi} = 1 - \dfrac{t}{\pi}$ ← arc length of "safe" area ← total circumference

$= \prod_{i=2}^{20} P(X_i \text{ not in } X_1 \text{ boundary}) = \boxed{\left(1 - \frac{t}{\pi}\right)^{19}}$

i.i.d

b) D = distance b/w X_1 and nearest point

Survival function of D:

$u = 1 - \frac{t}{\pi}$, $du = -\frac{1}{\pi} dt$, $dt = -\pi\, du$

$$E(D) = \int_0^{\pi} P(D > t)\, dt = \int_0^{\pi} \left(1 - \frac{t}{\pi}\right)^{19} dt = \int_{1-\frac{0}{\pi}=1}^{1-\frac{\pi}{\pi}=0} -\pi u^{19}\, dt = -\pi \int_1^0 u^{19}\, dt = -\frac{\pi}{20}\left(u^{20}\right)\Big|_1^0 = -\frac{\pi}{20}\left(-1^{20}\right)$$

$= \frac{\pi}{20}$ ← avg distance to nearest point

degrees: $\dfrac{\pi/20}{2\pi} = \dfrac{\pi}{20} \cdot \dfrac{1}{2\pi} = \dfrac{1}{40} \Rightarrow \dfrac{1}{40} \cdot 360° = \boxed{9°}$

total circumference

9. **Cancer screening (3 points).** A medical test has sensitivity 90% and false–positive rate 3%:

$$\mathbb{P}(T = 1 \mid C = 1) = 0.9, \qquad \mathbb{P}(T = 1 \mid C = 0) = 0.03.$$

Suppose the disease prevalence is very low, $\mathbb{P}(C = 1) = 0.001$. Compute the posterior probability of disease given a positive test:

$$\mathbb{P}(C = 1 \mid T = 1).$$

Bayes' Th$^{\underline{m}}$

$$P(C=1 \mid T=1) = \frac{P(T=1 \mid C=1)\,P(C=1)}{P(T=1)} = \frac{(0.9)(0.001)}{(0.9)(0.001) + (0.03)(1-0.001)}$$

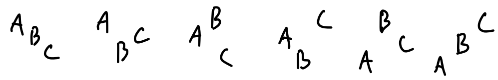
$$\hookrightarrow P(T=1) = P(T=1 \mid C=1)\,P(C=1) + P(T=1 \mid C=0)\,P(C=0) = 0.9(0.001) + 0.03(1-0.001)$$

Law of Total Probability

$$= \boxed{\frac{10}{343} \approx 0.02915 \ (2.915\%)}$$

10. **Follower Counts (3 points).** Suppose 420 people are sitting uniformly at random around a circle, each with a *distinct* number of TikTok followers. What is the expected number of people whose follower count is higher than both their immediate neighbors?

$$X_i = \begin{cases} 1, & \text{person } i \text{ has more followers than both neighbors} \\ 0, & \text{otherwise} \end{cases}$$



$$P(X_i = 1) = \frac{2}{3!} = \frac{1}{3}$$

$$X = X_1 + \cdots + X_{420}$$

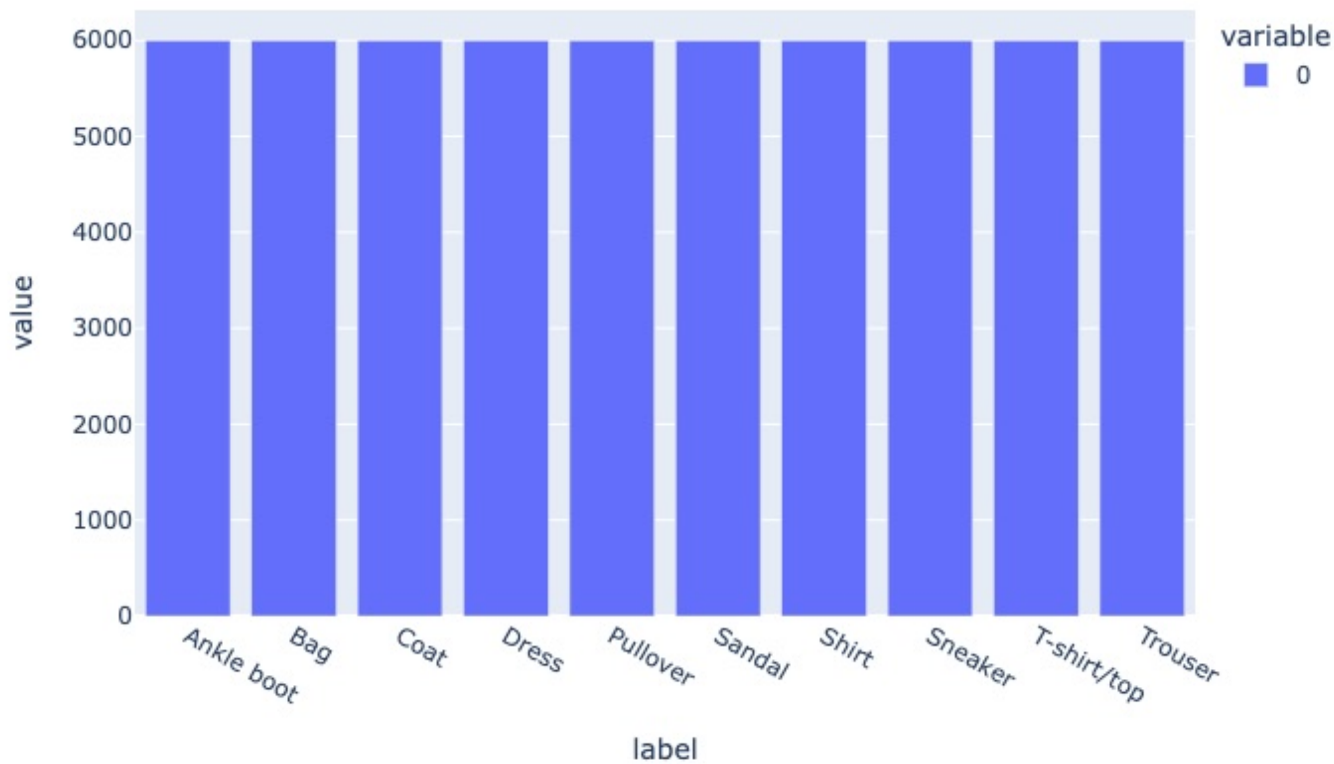
$$E(X) = \sum_{i=1}^{420} E(X_i = 1) = 420 \cdot E(X_1 = 1) = 420 \cdot P(X_1 = 1) = 420 \cdot \frac{1}{3} = 140$$

$$= \boxed{140 \text{ people}}$$

Coding

For every question labeled with **MANUAL** in the Coding part of this homework, please label and attach either your figures or your written answers below.

1c: Visualizing Label Distribution



2c: Visualizing Clusters

4g: Matrix Multiply Questions

1. The order of transformations does matter because matrix multiplication is non-commutative, so $AB \neq BA$. A good example of this would be a shift, then rotation vs rotation, then shift. The former would spin the image in a radius around the center, while the latter would first rotate about the center, then shift which is not the same.

2. Transformations such as shifting and rotation can be undone by multiplying the augmented matrix with the complement of those transformation matrices: $(-dx, -dy)$ for shifting and $(360 - \text{theta})$ for rotation. Blurring an image cannot be undone because the output is destructive since it replaces each pixel with the kernel mean of surrounding pixels, so the original pixel is lost in calculation.

3. Shifting and rotation (bilinear interpolation) can be undone, but data can be lost if shifting moves the image out of bounds such that an unshift cannot recover those lost pixels, or if we apply rotation without bilinear interpolation, since there are holes that may appear in which data is lost due to rounding.

4. Blurring cannot be undone with another matrix multiplication because the transformation replaces each pixel with the kernel mean of neighboring pixels, so the original pixel is lost. In other words, we cannot increase the resolution since the calculation in blurring is destructive.

4j: Analysis of Augmentation Techniques

Among all the augmentation techniques, the kernel blur augmentation performed the best. I believe the kernel blur performed so well because a small 3x3 kernel, while removing sharp edges and reducing the resolution, still maintained the same spatial location that the model can generalize. On the other end, the combined augmentation shift, rotate, blur performed the poorest. I believe this is due to the combo interfearing with the spatial weight of the model as well as rotation completely changing the axis of the entire image. It seems that the best transformations that the model can generalize are if the spatial boundary of the image remains intact.

5c: MAE, MSE, and R-Squared

MAE (mean absolute error) treats all errors linearly of which a single outlier does not disproportionately skew the metric. It is also better for interpretability, since you can look at the MAE and be able to know immediately how far on average your model is doing to the true dataset.

MSE (mean squared error) dramatically boosts large errors by squaring them which is an advantage if you want your model to be sensitive to big shifts in the data such that, during regression, your model can catch it. It is also differentiable everywhere, unlike MAE which is undefined at zero, so it is easier to optimize during GD.

R^2 measures the proportion of the variance in the dependent variable, or the y column, that is predictable from the independent variables, the x columns. R^2 is significant because it provides a universal scale from 0 to 1 that is dimensionless, allowing you to compare models across different datasets.

6d: Analysis of Class Accuracies and Confusion Matrix

A weird pattern that I noticed is that "Trouser" was disproportionately mispredicted over the other items. The highest mispredictions as shown in the confusion matrix are T-shirt/top and Coat.

Another pattern I saw was that "shirt" was being mispredicted quite frequently with a more even spread of mispredictions across the other classes and itself.

These patterns may suggest that the test set contains many new categories or is simply too different for our original MLP model to generalize. We may be dealing with underfitting our model.

8d: Comparing Data Augmentation and Rotation Correction

One reason the rotation correction struggles against data augmentation is that even when the MLPRegressor is off by a couple degrees, the correction "unrotation" transformation still rotates the original image into a position the final classifier has never seen before. Even though the error in the regressor model may be small, it could result in a massive disproportionate error in the classifier model.